

Siege on Olympus! — Deicides of the 7 Clay Prizes

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0.1 *A Formal Barrier Taxonomy for the Millennium Prize Problems in Lean 4*

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Document role: Self-contained research paper. Presents the machine-checked barrier taxonomy for all seven Clay Millennium Prize Problems: the `BarrierType` inductive, the `ym_is_unique_missing_foundation` theorem, the stacked/parallel sorry distinction, the vessel-content inseparability principle, the ZFC_t promotion gate inhabitants for RH and Hodge, the $P \neq NP$ structural coordinate theorem and dialethic resolution, and the primitive bridge connecting sorry boundaries to the Inscribing Grammar constraint algebra. Target venue: Journal of Formalized Reasoning / Journal of Automated Reasoning.

0.2 Two-Track Architecture

The Inscribing Grammar Lean library occupies two tracks: `Primitives/` (the 12-primitive constraint algebra) and `Inscribing/Millennium/` (the barrier taxonomy library documented here). This paper reports on the `Inscribing/Millennium/` track and its bridge to `Primitives/`.

Internal references within this paper use §N.

Library location: `MillenniumAnkh/Inscribing/Millennium/` — 43 modules. Build target: `lake build Inscribing`.

0.3 I. Introduction (v0.3.0, 2026-05-19)

The Clay Mathematics Institute seven Millennium Prize Problems have resisted proof for decades — and in one case, the Riemann Hypothesis, for over a century and a half. Proof assistants have formalized large bodies of mathematics, but existing efforts overwhelmingly target *results that are known*: the Last Theorem of Fermat, the Four Color Theorem, the Kepler Conjecture. This library formalizes *why specific problems occupy the positions they do* — the structural barriers that distinguish them from each other and from the solved.

The contribution is meta-level and computational: a machine-checked classification of proof obligations, the structural relationships between them, and their connection to the

Inscribing Grammar primitive constraint algebra. Every `sorry` is labeled by type; the type determines the research strategy for discharging it.

0.3.1 I.1 Why this matters

The distinction between barrier types has practical consequences for the formalization community.

MathlibGap sorries are actionable. A contributor with the right background can discharge them by formalizing a known proof. The `euler_opn_form` sorry in `OPN.lean` (Euler 1747) and the `mazur_torsion` sorry in `BSD.lean` (Mazur 1977) are of this type.

OpenProblem sorries define the research frontier. They are dischargeable only by solving the underlying mathematics. This distinction became clear during formalization of the BSD conjecture (§II.3), where the Mordell-Weil theorem was initially misassigned as an OpenProblem before its MathlibGap status was confirmed.

MissingFoundation sorries require building new mathematics before the question is fully stateable. An OpenProblem has a well-typed proposition whose truth value is unknown. A MissingFoundation sorry requires inhabiting a *type* that does not yet exist as a rigorous mathematical object. Yang-Mills is the only Millennium Problem in this category, and this uniqueness is a formal theorem (`decide`).

0.3.2 I.2 The sorry depth distinction

We introduce a formal notion of *sorry depth* distinguishing *stacked* from *parallel* proof obligations.

Stacked (Yang-Mills): sorry B depends on sorry A — the mass gap is not stateable as a proposition until the quantum Yang-Mills measure is known to exist. The foundation (`PathIntegralMeasure G`) is a prerequisite for defining the mass gap at all.

Parallel (BSD): three sorries are logically independent — Mordell-Weil (1922, MathlibGap), Mazur torsion (1977, MathlibGap), and the BSD rank formula (OpenProblem) — each dischargeable independently.

Both Yang-Mills and BSD have `sorryDepth = 2`. The structural difference is encoded in the barrier type.

0.3.3 I.3 Contributions

C1 — BarrierType taxonomy: A typed inductive with three constructors (`MathlibGap`, `OpenProblem`, `MissingFoundation`), formally distinct by `decide`, and computably assigned to all seven Millennium Problems.

C2 — `ym_is_unique_missing_foundation`: A theorem, proved by `decide`, that Yang-Mills is the only Millennium Problem whose barrier is `MissingFoundation`.

C3 — $P \neq NP$ structural coordinate theorem and dialethic resolution: `P_class_ne_NP_class` is proved in term mode: P has topology Θ_6 (`T_network`) and NP has topology $\Theta_\odot$ (`T_odot`); the distinct-constructor property of Lean’s inductive types closes the inequality. Primitive distance $d = 6.245$ across six dimensions $(\Theta, \check{R}, \mathcal{K}, \Gamma, \mathbb{I}, \Phi)$ places P and NP in structurally separated positions before any algorithm runs.

The dialethic layer: the Inscripting Grammar sits at $\Theta_\odot$ (self-referential closure requires $\Theta_\odot$) yet is distinct from `NP_class` on Dimensionality ($\check{\Theta}_\odot$ vs. $\check{\Theta}_\infty$). The `dialethic_resolution`

theorem packages all three: Grammar shares NP's topology AND $\text{Grammar} \neq \text{NP}$ AND $\text{P} \neq \text{NP}$. This conjunction is the O_∞ signature, not a contradiction: a system that inhabits $\Theta_\odot$ while being distinct from every specific $\Theta_\odot$ inhabitant is precisely what the self-modeling gate $\odot_{\bar{y}}$ requires. The grammar is the measure; P and NP are the measured.

C4 — NS critical Sobolev exponent: Machine-verified by `norm_num` that $0 < \frac{1}{2} < 1$ — the formal statement of why NS regularity is hard.

C5 — OPN in real Mathlib: the Touchard congruence type-checks using actual `Nat.Perfect` and `ArithmeticFunction.sigma`.

C6 — BSD in real Mathlib: `BSDRankConjecture` stated using actual `WeierstrassCurve` $\mathbb{Q}$ and `IsElliptic`; three parallel sorries formally justified.

C7 — Vessel-content inseparability (`VesselContent.lean`): The `Imscribing Grammar` provides two things simultaneously: the vessel (the crystal coordinate — the structural form of what a system is capable of being) and the content (the primitive algebra operations — what fills that vessel). These are constitutively inseparable.

Formalized in three theorems. `form_uniqueness`: every inscribable system has exactly one crystal coordinate ($\exists! c, \text{Imscribes } M \ c$). `content_containment`: the vessel constrains the content ($\text{Imscribes } M \ c \rightarrow \text{Reachable } M \ r \rightarrow \text{WithinAlgebra } c \ r$). `vessel_fills_itself`: the biconditional $\text{Reachable } M \ r \iff \text{WithinAlgebra } c \ r$.

The biconditional is the non-trivial claim. `form_uniqueness` alone is a labeling property: knowing the unique coordinate of M no more determines M 's behavior than knowing a name determines speech. `vessel_fills_itself` asserts structural constraint in both directions: the vessel bounds the content ($\rightarrow$) and the content exhausts the vessel ($\leftarrow$) — no empty room.

C8 — ZFC_t gate inhabitants for RH (`RH_GateInhabitants.lean`): All four ZFC_t promotion slots for the Riemann Hypothesis are inhabited with concrete structures. `FrobeniusZeroSymmetry`: $\theta = (1 - \cdot)$ with involution proved by `ring`. `FunctionalEquationDual`: $s \mapsto 1 - s$ preserves $\text{Re}(s) = \frac{1}{2}$ on the critical line. `ZFunctionWinding`: $N(T)$ counting function (Hardy 1914). `PrimeZeroBridge`: $\psi(x) = x$ explicit formula placeholder. `ZFCt_RHCertificate` assembles all four.

Key theorems without sorry: `frob_gate_without_forcing` ($\theta(s) = s \iff s = \frac{1}{2}$, proved by `linear_combination`); `rh_forcing_implies_rh` (`RH_ForcingTheorem` $\rightarrow$ `RiemannHypothesis`, proved by `rw [rh_barrier]` then curried introduction). The gate structures are inhabited and the fixed locus is characterized; `RH_ForcingTheorem` — that all nontrivial zeros lie in that fixed locus — is the single open gap.

C9 — ZFC_t gate inhabitants for Hodge (`Hodge_GateInhabitants.lean`): `HodgeLRDual` inhabits the `LR_DUAL` slot (Hodge decomposition $H^{p,q} \leftrightarrow H^{q,p}$ via σ). `HodgePM_Z2` inhabits the `PM_Z2` slot (complex conjugation as $\mathbb{Z}_2$ involution, $\sigma^2 = \text{id}$, fixed locus $H^{p,p}$). `HodgeWinding` inhabits the `ZWIND` slot (Griffiths group rank function; Griffiths 1969 showed this is non-trivial for $p \geq 2$).

Key theorem: `hodge_forcing_equiv_hodge` (`Hodge_ForcingTheorem` $\iff$ `HodgeConjecture`), proved in term mode via `hodge_sorry_requires_cycle_class_surjectivity.symm`. Structural parallel with RH: both have a $\mathbb{Z}_2$ involution, an identified fixed locus, and

an open forcing claim. The substrate differs — $\mathbb{C}$ (RH) vs. $H^{p,q}(X, \mathbb{C})$ (Hodge) — the pattern is the same.

C10 — PrimitiveBridge.lean: Formal connection between sorry boundaries and primitive field transitions in the Inscribing Grammar; `BarrierPrimitiveCertificate` structure; `primitive_bridge_master` theorem.

C11 — RH–Lee–Yang structural correspondence: Machine-checked theorem that the Riemann ζ zeros and Lee–Yang partition-function zeros share the Criticality assignment `Phi_c_complex`. Structural distance 1.0 (machine-checked) identifies the polarity primitive $\mathcal{P}$ ($\mathcal{P}_{\text{sym}}$ vs. $\mathcal{P}_{\pm}^{\text{sym}}$) as the essential structural gap between the two zero distributions.